

Databases

Labwork 2

Task 1

The initial table Booking_receipt contains information about a booking, passenger, flight, airports, airline and seats. The column seat_numbers can contain several values in one field, for example "12A, 12B".

This violates First Normal Form (1NF), because every field must contain one atomic value. The responsible attribute is seat_numbers.

Task 2

Step 1 — First Normal Form (1NF)

The seat list is split so that each row contains one seat number.

```
Booking_receipt(booking_id, passenger_full_name, passenger_passport_number, flight_number,  
departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name,  
seat_number, ticket_price)
```

Primary key: (booking_id, seat_number)

Now each value is atomic, so the relation satisfies 1NF.

Step 2 — Second Normal Form (2NF)

The 1NF table has a composite key (booking_id, seat_number). Passenger and booking information depends only on booking_id, so there are partial dependencies. The relation is split into Booking and Booking_Seat.

```
Booking(booking_id, passenger_full_name, passenger_passport_number, flight_number,
departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name,
ticket_price)
```

Primary key: booking_id

```
Booking_Seat(booking_id, seat_number)
```

Primary key: (booking_id, seat_number); booking_id is a foreign key.

Step 3 — Third Normal Form (3NF)

Transitive dependencies are removed. For example, booking_id determines flight_number, while flight_number determines airline_name. Airport and airline information are therefore separated from Booking.

```
Booking(booking_id, passenger_full_name, passenger_passport_number, flight_number, ticket_price)
```

```
Booking_Seat(booking_id, seat_number)
```

```
Flight(flight_number, departure_airport_name, arrival_airport_name, airline_name)
```

```
Airport(airport_name, city)
```

```
Airline(airline_name)
```

Task 3

Functional dependencies used in normalization:

```
booking_id → passenger_full_name
booking_id → passenger_passport_number
booking_id → flight_number
booking_id → ticket_price
flight_number → airline_name
flight_number → departure_airport_name
flight_number → arrival_airport_name
departure_airport_name → departure_city
arrival_airport_name → arrival_city
```

Task 4

The current primary key of Flights is flight_id. The flight attributes depend on flight_id. Airline and airport descriptive information is kept in separate tables and referenced through foreign keys.

The current Flights table therefore avoids storing airline and airport descriptive attributes directly in the flight row. The two airport references are departing_airport_id and arriving_airport_id, while airline_id references the Airline table.

DDL — Database Tables

The database contains the ten required tables.

- Sequences
- ✓  Tables (10)
 - >  airline
 - >  airport
 - >  baggage
 - >  baggage_check
 - >  boarding_pass
 - >  booking
 - >  booking_flight
 - >  flights
 - >  passengers
 - >  security_check

DDL — Airline

Airline table and its six columns.

▼  airline

▼  Columns (6)

 airline_id integer

 airline_code character varying(30)

 airline_name character varying(50)

 airline_country character varying(50)

 created_at timestamp without time zone

 updated_at timestamp without time zone

DDL — Airport

Airport table and its seven columns.

▼  airport

▼  Columns (7) 

 airport_id integer

 airport_name character varying(50)

 country character varying(50)

 state character varying(50)

 city character varying(50)

 created_at timestamp without time zone

 updated_at timestamp without time zone

DDL — Baggage

Baggage table and its five columns.

▼  baggage

▼  Columns (5)

 baggage_id integer

 weight_in_kg numeric(4,2)

 created_at timestamp without time zone

 updated_at timestamp without time zone

 booking_id integer

DDL — Baggage_check

Baggage_check table and its six columns.

▼  baggage_check

▼  Columns (6)

 baggage_check_id integer

 check_result character varying(50)

 created_at timestamp without time zone

 updated_at timestamp without time zone

 booking_id integer

 passenger_id integer

DDL — Boarding_pass

Boarding_pass table and its six columns.

▼  boarding_pass
▼  Columns (6)
 boarding_pass_id integer
 booking_id integer
 seat character varying(50)
 boarding_time timestamp without time zone
 created_at timestamp without time zone
 updated_at timestamp without time zone

DDL — Booking

Booking table. The final column is ticket_price.

▼  booking

▼  Columns (8)

 booking_id integer

 flight_id integer

 passenger_id integer

 booking_platform character varying(50)

 created_at timestamp without time zone

 updated_at timestamp without time zone

 status character varying(50)

 ticket_price numeric(7,2)

DDL — Booking_flight

Booking_flight table and its five columns.

▼  booking_flight
▼  Columns (5)
 booking_flight_id integer
 booking_id integer
 flight_id integer
 created_at timestamp without time zone
 updated_at timestamp without time zone

DDL — Flights

Flights table. departing_gate is shown as text.

▼  flights

▼  Columns (12)

-  flight_id integer
-  sch_departure_time timestamp without time zone
-  sch_arrival_time timestamp without time zone
-  departing_airport_id integer
-  arriving_airport_id integer
-  departing_gate text
-  arriving_gate character varying(50)
-  airline_id integer
-  act_departure_time timestamp without time zone
-  act_arrival_time timestamp without time zone
-  created_at timestamp without time zone
-  updated_at timestamp without time zone

DDL — Passengers and Security_check

Passengers table.

- passengers
 - Columns (10)
 - passenger_id integer
 - first_name character varying(50)
 - last_name character varying(50)
 - date_of_birth date
 - gender character varying(50)
 - country_of_citizenship character varying(50)
 - country_of_residence character varying(50)
 - passport_number character varying(20)
 - created_at timestamp without time zone
 - updated_at timestamp without time zone

Security_check table.

- ▼  security_check
 - ▼  Columns (5)
 -  security_check_id integer
 -  check_result character varying(20)
 -  created_at timestamp without time zone
 -  updated_at timestamp without time zone
 -  passenger_id integer

Final query result: flights joined with airlines and departure/arrival airports.

Data Output

Messages

Notifications

+

📄

▼

📋

▼

🗑️

🗑️

📁

⬇️

📈

SQL

Showing rows: 1 to 5

🔍

Page No: 1 of 1

⏪ ⏩

	flight_id integer	airline_name character varying (50)	departure_airport character varying (50)	arrival_airport character varying (50)	departing_gate text	arriving_gate character varying (50)
1	1	Air Astana	Almaty International Airp...	Astana International Airp...	A1	B1
2	2	Turkish Airlines	Astana International Airp...	Istanbul Airport	A2	B2
3	3	Emirates	Istanbul Airport	Dubai International Airport	A3	B3
4	4	Lufthansa	Dubai International Airport	Frankfurt Airport	A4	B4
5	5	FlyArystan	Frankfurt Airport	Almaty International Airp...	A5	B5